

Small-Scale Text Diffusion for a Psychology Assistant: A Discrete-Diffusion Pipeline, a Blind-Annotation Retraction, and an Honest Data-Bound Scaling Null

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Abstract

We build an English psychology / mental-health text-generation model on a single consumer GPU and report the full chain of results — several of them negative — with deliberate measurement rigor. The methodological spine is a self-caught error: an effect we first “found” non-blind (corpus cleaning halving confabulation, 57% $\rightarrow$ 29%) *vanished under blind annotation* (Fisher $p = 1.0$); we retract it and treat the caught confirmation bias, and the blind protocol that exposed it, as a primary contribution. Substantive findings: (1) **Architecture** — continuous (Gaussian) text diffusion at 60M parameters / 2.56B tokens produces topic-coherent *word salad*, while discrete masked diffusion (MDLM) at the same scale yields grammatical sentences. (2) **Decoding** — the discrete model’s dominant failure, repetition collapse, is a *decode artifact, not a capacity limit*: confidence-based unmasking plus repetition controls remove it with no weight change (rep-1 -53% at fixed temperature). (3) **Data** — a strict-filtered 0.77B-token corpus is 3.3 $\times$ cheaper than the 2.56B noisy corpus and no worse on the most sensitive eval (paired cosine $d = +0.0094$, win-rate 0.527): cleaning buys *cost*, not quality. (4) **Scale** — doubling to 120M parameters at fixed data moves neither aggregate quality (paired $d = -0.0094$, win-rate 0.479) nor the blind homonym-drift floor (Fisher $p = 0.77$); a measured null we attribute to an *inferred* data-bound regime (6.4 vs. 11.6 tokens/param against a ≈ 20 optimum), falsifiable only by a data-fed rerun. The shipped model is the cheaper 66.8M clean checkpoint, deployed fully on-device (Android, ONNX, fp32). A closing audit re-derived every headline statistic from raw artifacts (Appendix A).

1 Introduction

The goal was a compact diffusion language model specialized to psychology and mental-health dialogue, trainable and deployable on one RTX 3090. Diffusion LMs promise controllable, non-autoregressive generation, but little is documented about what works at the small scale a hobbyist or edge budget allows. This paper is an engineering-and-science log: what failed, what worked, and where our own measurements misled us until corrected.

The project ran as seven stages. For each, the body states *what was done*, *why*, and *what it led to* — the causal chain matters, because three pivots (architecture switch, sampler rewrite, corpus rebuild) were each forced by a documented negative result, and one headline “positive” was retracted on its own evidence:

1. **Data + tokenizer** (§2): 2.56B-token psychology-dominant corpus, 261k-pair SFT set, fixed 16k SentencePiece vocab → one tokenizer and recipe frozen for every later arm.
2. **Continuous diffusion, v1–v3** (§3): topic control emerged; grammar never did; five extra sampler tricks made outputs strictly worse → pivot to discrete diffusion at identical budget.
3. **Discrete MDLM** (§4): same scale, grammatical English → pipeline gate opened.
4. **Decode fix** (§5): repetition collapse removed with zero retraining → collapse = decode order, sampler frozen for all evals and deployment.
5. **Clean corpus + factorial** (§6–7): strict filter to 0.774B tokens; non-blind “confab halved” *retracted* under blind labels → standing blind protocol.
6. **Data and scale axes** (§8–9): paired 1000-prompt eval + blind traps → cleaning buys cost not quality; 2× params at fixed data is a null.
7. **Deployment** (§10): 66.8M-clean exported to ONNX, runs on-device.

Related work. The discrete model follows masked-diffusion LMs — MDLM [1], SEDD [2] — with confidence-ordered unmasking from MaskGIT [3]; the continuous baseline is Diffusion-LM [4] with DiffuSeq-style conditioned SFT [5]; the backbone is a DiT [9]. Scaling is read through Chinchilla [6]; the data result confirms, in the diffusion regime, the curated-small-data line of TinyStories [7] and phi [8]. The combination — masked diffusion, single GPU, domain-specialized, with a blind-annotation retraction — is, to our knowledge, not covered by these.

2 Setup: model, objective, data

Backbone. Diffusion Transformer with AdaLN-zero timestep conditioning, RMSNorm, SwiGLU MLP, SDPA attention, learned positions, tied output head ($\text{logits} = hE^\top + b$), sequence length 512. Two sizes (Table 1); 120M scales width and depth at fixed head dim 64. Training is pure bf16 — parameters, activations, gradients; fp32 only inside RMSNorm, the sinusoidal time embedding, and loss reductions — with AdamW8bit [10], no fp32 master weights. This single-card regime is held constant across all arms, so it is not a confound for any comparison.

Table 1: Architectures. Parameter counts are from training-log prints, not estimates.

name	hidden	layers	heads	MLP inter	params	tok/param @0.774B
66.8M	512	12	8	1408	66.8M	11.6
120M	640	15	10	1664	121.4M	6.4

Objective. MDLM [1]: per sequence draw $t \sim U(0.01, 0.99)$, mask each token independently with probability t (vocab 16,000 + 1 [MASK]), cross-entropy on masked positions only. SFT is DiffuSeq-style: only response positions are noised, prompts stay clean as conditioning; zero new parameters, pretrain checkpoint loaded `strict=True`.

Recipe (frozen). Batch $32 \times \text{accum } 8$ ($= 256$ seqs $= 131\text{k}$ tokens/step), LR $4 \cdot 10^{-4}$ cosine, 2% warmup, weight decay 0.1, betas (0.9, 0.99), clip 1.0; pretrain 20k steps ($\approx 2.62\text{B}$ seen tokens ≈ 1 epoch noisy / ≈ 3.4 epochs clean), SFT 8k. Wall-clock per run on the 3090: $\approx 9.3\text{h}$ pretrain + 3.7–3.85h SFT at 66.8M; $\approx 26.5\text{h} + 5.8\text{h}$ at 120M (run ledger: Appendix D).

Data. Pretrain corpus, psychology-dominant English, two versions: **noisy** 2,559,832,146 tokens / 2,163,570 docs ($\approx 69\%$ loosely psych-filtered Pile, plus peS2o, OpenAlex psychology, PMC, Gutenberg, PsyArXiv, and a small RU $\rightarrow$ EN translated set); **clean** 773,925,936 tokens / 832,822 docs (§6). SFT: 261,443 (prompt, response) pairs at length 512 (assistant + mental-health + translated psychology dialogue), shared by every arm.

What it led to. Every later contrast changes one factor on identical infrastructure. The one violation of this rule was the non-blind labeling episode (§7) — caught and retracted.

3 Stage 1 — continuous diffusion fails, three samplers deep

We first built a continuous, x_0 -parameterized Diffusion-LM-style model: 60.8M params, 2.56B tokens, DDIM, classifier-free guidance, self-conditioning.

v1–v2. CFG controls topic: at $\text{cfg} \approx 2.5$ outputs lock into a therapy register (on-topic content-word density $\sim 10\% \rightarrow \sim 35\%$), and constraints/clamp suppress garbage. Grammar never emerges — outputs are word salad in the right vocabulary; CBT-vs-ACT and Vygotsky prompts fail outright. **v3 (negative).** Five further techniques (CFG cosine ramp, stochastic DDIM, final top- p /temp, iterative refinement, short max-length) were ablated on the same checkpoint: together they made outputs strictly worse — register lock-in $3/5 \rightarrow 0/5$, on-topic density $\sim 35\% \rightarrow \sim 15\%$; final top- p over 16k SPM occasionally emits rare-Unicode one-shots.

What it led to. Inference-time engineering has a ceiling and we hit it: the limit is architectural. The arm survives only as a matched-budget baseline.

4 Stage 2 — discrete MDLM gives grammar at the same scale

The 66.8M pretrain reached masked-CE $9.80 \rightarrow 4.28$ (token acc 29.9%) in 20k steps; SFT $4.80 \rightarrow 3.71$ (35.8%). Coherent clauses appeared at 20% of training; post-SFT, coping/advice prompts produce grammatical multi-sentence on-topic English — the failure mode shifted to repetition collapse on factual/homonym prompts, expected at 60M.

What it led to. The architecture finding (discrete $\gg$ continuous at small scale, holding scale/data/tokenizer fixed), a frozen pipeline for matched experiments, and a defined next target: collapse.

5 Stage 3 — repetition collapse is decode-bound

Replacing random-order unmasking with confidence-ordered unmasking (cosine schedule) plus repetition penalty 1.3, neighbor-ban 2, no-repeat-3-gram fixes the same checkpoint, no retraining (Table 2).

Table 2: v1 vs. v2 sampler, identical 66.8M-noisy weights, 5 matched prompts, temp fixed 0.9.

metric	v1 random	v2 conf+rep	Δ
distinct-1	0.647	0.835	+29%
distinct-2	0.929	0.971	+0.042
rep-1	0.353	0.165	−53%
max-token-freq	0.090	0.064	−29%

A first draft compared v1 at temp $\{0.7, 1.0\}$ to v2 at 0.9 and showed rep-1 -62% ; rerun at fixed 0.9 the effect attenuated to -53% — part of the original delta *was* temperature, and the controlled number is reported. **What it led to:** collapse = decode order, not capacity; sampler frozen for all evals (and the phone). The decode fix did *not* reduce homonym drift, isolating it as a data/capacity question.

6 Stage 4 — strict-filtered clean corpus

Drift such as *Vygotsky* $\rightarrow$ “coastal/coral” implicated the Pile component. Strict filter — keep iff core-psych hits ≥ 3 and core $\geq$ off-domain — yields 0.774B tokens (30% of noisy) at far higher purity; the rule *never weaken the filter to buy tokens* held, accepting under-feeding of 120M. A pretrain-level probe (clean stays developmental on *proximal/distal*; noisy drifts to islands) motivated — but does not establish — a corpus effect; that test is §7. **Led to:** the corpus axis and the $3\times$ cheaper production model.

7 Stage 5 — the retracted finding and the blind protocol

Factorial: decode (v1/v2) $\times$ corpus (noisy/clean) at 66.8M. Decode reproduced on both corpora. On 14 homonym traps $\times$ 2 seeds the author labeled drift *knowing condition and hypothesis*: 16/28 vs 8/28 — “cleaning halves confabulation”. Three checks before publishing: (1) **CI**s: Wilson $[0.39, 0.73]$ vs $[0.15, 0.47]$, Fisher $p = 0.058$ — not significant. (2) **Blind** (decisive): 56 pooled, shuffled, source-stripped, two fresh annotators:

annotator	noisy	clean	Fisher p
author, non-blind	16/28 (57%)	8/28 (29%)	0.058
A, blind	11/28 (39%)	10/28 (36%)	1.000
B, blind	7/28 (25%)	7/28 (25%)	1.000

$\kappa_{AB} = 0.71$: criterion reliable, effect absent — the non-blind delta was confirmation bias. (3) **Specificity**: clean is not blander (TTR 0.899 vs 0.900; content-word 0.556 vs 0.562; hapax 0.910 vs 0.907).

Retraction; reading. Trap drift splits $\sim 14/16$ homonym-type (*drive* $\rightarrow$ vehicles) — corpus-invariant lexical floor — vs $\sim 2/16$ contamination-type, all cleaning can fix. **Led to:** standing protocol — pool, shuffle, strip, ≥ 2 fresh blind annotators, decode after, report κ /CI/Fisher — used throughout §9.

8 Stage 6a — data axis: cleaning buys cost, not quality

Same arch/SFT; pretrain clean 0.774B vs noisy 2.56B. Ceiling metrics: tie (above). Degeneration, 1000 prompts: distinct-2 0.993/0.991, rep-1 0.137/0.141, zero collapse. Sensitive eval, MiniLM cosine-to-gold on 725 paired prompts: 0.500 vs 0.490, $d = +0.0094$, CI $[+0.0003, +0.0186]$, $t(724) = 2.01$, win 0.527 — borderline significant, negligible size. **Led to:** cleaning’s benefit is cost ($3.3\times$ fewer tokens, no penalty); confirms TinyStories/phi in diffusion. Caveats: volume/purity confounded (noisy-0.77B not run); slim fluency edge $\neq$ trap result (§7); ± 0.0094 symmetry with §9 shows the metric resolves 0.01 but signs are single-seed.

9 Stage 6b — scale axis: a data-bound null

120M on identical clean data (transient CUDA fault absorbed by auto-resume). Pre-registered evals: **Aggregate paired:** 0.490 vs 0.500, $d = -0.0094$, CI $[-0.0188, -0.0001]$, $t = -1.96$, win 0.479 — practical tie. **Blind traps** (3 fresh annotators, κ 0.767/0.867/0.825): 120M 7/28 vs 66.8M 9/28, Newcombe $[-0.294, +0.161]$, Fisher $p = 0.768$ — floor 25–32% both. **Led to:** measured null; inferred data-bound mechanism (6.4 vs 11.6 tok/param, optimum ≈ 20); production = 66.8M-clean. Scope: single seed/cell, two checkpoints, in-distribution golds; falsifier = 2.4B-token 120M.

10 Stage 7 — on-device deployment; conclusions

66.8M-clean exported to ONNX (286 MB asset; debug APK 365 MB), ONNX Runtime Mobile CPU, fp32 inference of bf16 weights (no quantization), v2 sampler ported, diffusion visualized (Figure 1). **Conclusions:** discrete > continuous at small scale; collapse is a decode problem; cleaning buys cost; doubling params at fixed data buys nothing measured; *blind annotation is non-negotiable*.

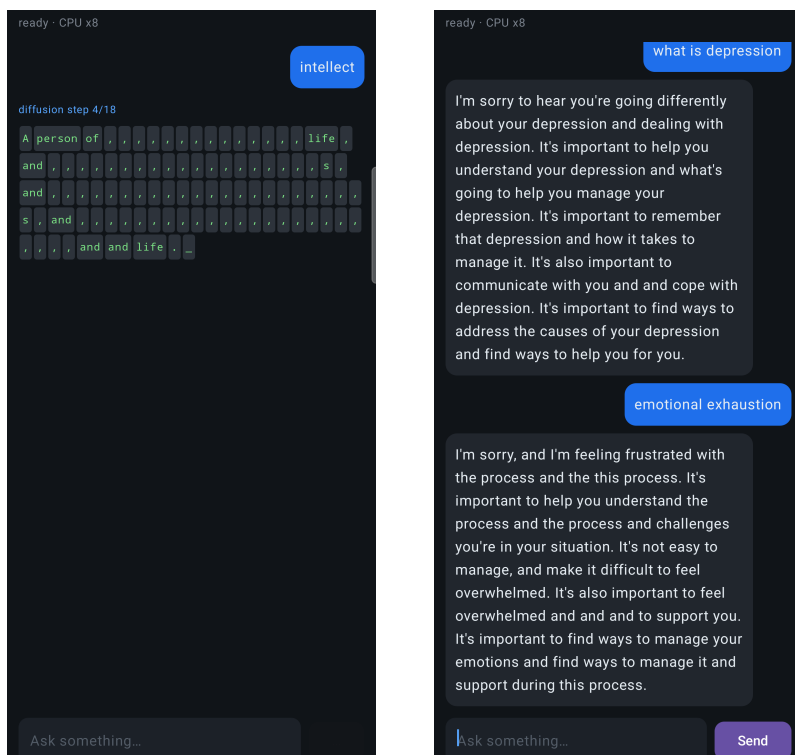


Figure 1: The 66.8M-clean model on a phone (ONNX Runtime Mobile, 8 CPU threads). Left: masked-diffusion decoding mid-generation (step 4/18) — unmasked tokens appear in confidence order. Right: completed chat answers with the §5 sampler (steps 28, temp 0.9, rep 1.3). Outputs are grammatical, on-topic, mildly repetitive — consistent with the limitations stated for this scale.

A Audit map: every headline number, claim → artifact → status

A closing audit (2026-06-10) re-derived each headline statistic from raw on-disk artifacts rather than from the working notes. Counts were verified against `meta.json` and byte sizes; the blind-trap statistics were recomputed by rerunning `eval/score_blind_traps.py` over the archived key and

annotator files; the paired cosines were recomputed end-to-end (re-embedding all outputs) with `eval/paired_sig.py`.

Table 3: Claim $\rightarrow$ artifact $\rightarrow$ recomputed value. Paths relative to `raw_data/004_train/`.

claim	artifact	recomputed
noisy corpus 2.56B tok, 2.16M docs	<code>pretrain_en_diff_clean/meta.json</code>	2,559,832,146; verified
clean corpus 0.774B tok, 833k docs	<code>pretrain_en_clean_v2/meta.json</code>	773,925,936; verified
SFT 261,443 pairs $\times$ 512 params 66.8M / 121.4M	<code>sft_en/meta.json</code> , bin size	exact; verified
runs 20k pre, 8k SFT, $\times$ 6	run logs	verified
decode fix (Table 2)	<code>six mdlm_*.log</code> , 05-29..31	steps and dates verified
retraction p values	<code>RESULTS_matched...md</code>	verified (temp-controlled)
data axis $d = +0.0094$	56-sample key, counts	0.058 / 1.000 / 1.000; verified
scale axis $d = -0.0094$	725 pairs re-embedded	[+0.0003, +0.0186], $t=2.01$, win 0.527; verified
blind scale traps	725 pairs re-embedded	[-0.0188, -0.0001], $t = -1.96$, win 0.479; verified
1000-Q set, gold pairs	<code>annot_{A,B,C}.txt</code> , key	7/28 vs 9/28, $p=0.768$, κ .77/.87/.83; verified
ONNX asset “ $\approx$ 243 MB”	line counts	1000, 725; verified
	app assets dir	286.4 MB; corrected

Archival notes: the §7 blind labels survive as counts in the results file (per-sample files were kept only for the scale round, which fully recomputes); the *noisy* dataset lives in the historically named directory `pretrain_en_diff_clean` (the clean one is `pretrain_en_clean_v2`); Wilson upper bound for 16/28 rounds to 0.73 (notes printed 0.74).

B Blind annotation protocol (standing)

(1) Generate condition pools (here 2 seeds $\times$ 14 traps per checkpoint). (2) Pool, shuffle, strip identifiers; assign neutral IDs (`build_blind_traps*.py`). (3) ≥ 2 (latterly 3) fresh annotators — LLM sub-agents never exposed to source, hypothesis, or direction — label drift yes/no per item; criterion only. (4) Decode by key; report per-source rates, Wilson 95% CIs, Newcombe diff CI, two-sided Fisher exact, pairwise Cohen κ (`score_blind_traps.py`). (5) A non-blind label may motivate, never headline. Triggered: the author 57% vs 29% became blind $p = 1.0$.

C Verbatim prompts given to the LLM sub-agents (Claude)

All annotation and curation labor besides the author was performed by fresh Claude sub-agents (Claude Code **Agent** tool, general-purpose, one task per agent, no memory of the project, no access to hypotheses or checkpoint identities).

Blind drift annotators (§7, §9). Each annotator received only the pooled, shuffled, source-stripped sample file (`blind_traps*.txt`); its header is the entire task specification. Scale round, verbatim (the §7 round was identical except the domain list lacked physics/cosmology and economics):

Label each item DRIFT=yes/no. DRIFT=yes if a substantial part of the response is about a NON-psychology domain (physics/cosmology, marine biology,

automotive, military, pure statistics/math, programming, geometry, economics, anatomy-as-physiology, etc). DRIFT=no if it stays within psychology/therapy/cognition/development EVEN IF factually wrong, vague, or repetitive. Repetition or incoherence alone is NOT drift.

Annotators returned one Sxxx: yes/no line per item (files annot_A-C.txt); decoding against the key happened only afterwards.

1000-question curators (§8–9). Candidate questions were machine-extracted in-distribution from the three SFT sources, psych-filtered, deduplicated, and split into 20 shards of 150; each of 20 sub-agents was instructed to curate, from its shard, the best 50 distinct well-formed English psychology questions, with a further 8 agents reconciling overlaps to exactly 1000; gold answers were recovered for 725 by matching SFT targets. Scripts: `build_q_shards.py`, `recover_answers.py`. Curators never saw model outputs, so the prompt set predates and is independent of every model under comparison.

D Run ledger (verified from logs)

run	steps	wall	checkpoint
mdlm_v1 (noisy pre)	20k	05-29 00:47–10:05	mdlm_v1/final
mdlm_sft (noisy)	8k	10:06–13:48	mdlm_sft/final
mdlm_v1_clean	20k	15:30–00:45	mdlm_v1_clean/final
mdlm_sft_clean (prod)	8k	00:49–04:38	mdlm_sft_clean/final
mdlm_120m	20k	05-29–05-30 21:47	mdlm_120m/final
mdlm_sft_120m	8k	21:50–03:39	mdlm_sft_120m/final

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